

5G NR L2/L3 + Web Dev Interview Notes

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1 5G NR L2/L3 Protocols

5G NR L2/L3 handles signaling, QoS, reliability and scheduling between UE and gNB.

1.1 RRC (L3) - Key Functions

- Broadcasts MIB/SIBs for system info
- Manages RRC states: IDLE/INACTIVE/CONNECTED
- Connection setup/reconfig/release
- Security activation, bearer control
- Handover/measurement configuration

1.2 RLC (L2) - Transfer Modes

- **TM:** No header/ARQ (control channels)
- **UM:** Unacked, low latency, reordering
- **AM:** ARQ retransmissions, status reports
- Segmentation/concatenation/reassembly

1.3 SDAP (L2) - QoS Mapping

- Maps QoS flows → DRBs using QFI
- Reflective QoS (UL mirrors DL rules)
- 1:1 or N:1 flow-to-bearer mapping

1.4 MAC (L2) - Scheduling

- Logical → transport channel mapping
- Multiplexing, HARQ coordination
- Buffer status reports, scheduling requests

1.5 Advanced MAC Features

- Proportional fair/QoS-aware scheduling
- Configured grant (Type 1/2) for URLLC
- Carrier aggregation, DRX power saving
- Beam failure recovery

2 Web Development Interview Prep

2.1 HTML/CSS Essentials

- Semantic tags: `<header>`, `<article>`, `<nav>`
- Flexbox: `justify-content`, `align-items`
- CSS Grid: `grid-template-areas`
- Media queries: `@media (max-width: 768px)`
- Box model: `padding`, `border`, `margin`

2.2 JavaScript Core

- `let/const/var` scoping differences
- Arrow functions: `() => {}`
- Array methods: `map/filter/reduce`
- Promises: `.then().catch()`, `async/await`
- Event handling: `addEventListener`
- Closures, `this` binding

2.3 SQL Fundamentals

- JOIN types: INNER, LEFT, RIGHT, FULL
- GROUP BY + HAVING vs WHERE
- Subqueries, CTEs: WITH clause
- Window functions: ROW_NUMBER(), RANK()
- Indexes: B-tree vs hash, composite keys
- EXPLAIN for query optimization